DESIGN AND SIMULATION OF A SECURE AND SCALABLE ENTERPRISE NETWORK USING VLAN SEGMENTATION AND ACCESS CONTROL

A Senior Project Report

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By

Raul Lozoya

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Table of Contents

I. Abstract

My project presents the design and simulation of a secure and scalable enterprise network using Virtual Local Area Network (VLAN) segmentation and access control mechanisms. Modern organizations depend on reliable and secure network infrastructures to support communication and data exchange across multiple departments. However, poorly designed networks can lead to security vulnerabilities, inefficient traffic management, and increased operational risks.

The proposed network architecture follows established enterprise design principles and incorporates VLAN based segmentation to logically separated departments within the organization. Inter VLAN communication is controlled using routing and Access Control lists (ACLs) to enforce security policies and restrict unauthorized access. The network is implemented and tested using Cisco Packet Tracer, allowing evaluation of connectivity, performance, and security within a simulated environment.

By integrating structured design methodologies, logical segmentation, and network-level security controls, this project demonstrates a cost-effective and practical approach to enterprise network development. The simulation validates the effectiveness of the proposed architecture and highlights the importance of secure network design in modern enterprise environments.

II. Introduction

1.1 Background and Significance

Modern organizations rely heavily on computer networks to support daily operations, communications, and data management. As enterprises grow, their networks become increasingly complex and vulnerable to security threats such as unauthorized access or data breaches. Poorly designed networks that lack segmentation and access control often expose sensitive organizational data to unnecessary risk.

This project is significant because it demonstrates how a secure enterprise network can be designed and tested using simulation tools rather than expensive physical infrastructure. By using Cisco Packet Tracer, the project provides a cost-effective and practical approach to modeling real-world enterprise networks while applying industry-standards network practices.

Enterprise network design plays a crucial role in ensuring secure, reliable, and efficient communication between different departments within an organization. Industry best practices emphasize the use of logical network segmentation, controlled routing, and security policies to protect sensitive data while maintaining performance. Technologies such as Virtual Local Area Networks (VLANs) and Access Control Lists (ACLs) are widely used to enforce these principles.

The significance of this project lies in demonstrating how a secure and scalable enterprise network can be designed and evaluated without the need for costly physical hardware. By using network simulation tools, organizations and students can model real-world scenarios, test security policies, and identify weaknesses before deployment. This approach provides a practical and cost-effective solution for understanding and applying enterprise networking concepts.

1.2 Project Description

The objective of this project is to design and simulate a secure and scalable enterprise network that supports multiple departments with different access requirements. The network will be logically segmented using VLANs and inter-VLAN communication will be controlled using routing and access control mechanisms.

The project will include the design of an IP addressing scheme of how implementation of routing protocols and configuration of security features will be validated through network testing. The simulation will allow evaluation of network performance, connectivity, and security without requiring physical networking hardware.

The project involves creating a structured IP address scheme, configuring routing between VLANs, and implementing security controls using Access Control lists to restrict unauthorized communication. The network will be simulated using Cisco Packet Tracer, allowing for testing of connectivity, performance, and security policies in a controlled environment. Through simulation and testing, the project aims to evaluate how enterprise networking techniques improve security, scalability, and overall network efficiency.

1.3 Target Audience

The intended audience for this project includes network administrators and IT professionals seeking practical examples of secure enterprise network design and implementation. Small and medium-sized organizations may also benefit from this project as it demonstrates how cost effective network segmentation and security controls can be implemented without the need for expensive physical infrastructure. Additionally, Computer Technology and Information Technology students can use this project as a learning reference for understanding VLAN segmentation, inter-VLAN routing, access control mechanisms, and network simulation. The report is also intended to support academic evaluators by clearly documenting the design decisions, configuration steps, testing procedures, and overall validation of the proposed network architecture.

2. Prior Related Work (Literature Survey)

2.1 Enterprise Network Design Models

Previous studies and industry guidelines emphasize the importance of hierarchical network design models typically consisting of core, distribution, and access layers. These models improve scalability, performance, and fault tolerance in enterprise networks. Cisco's enterprise network design framework is widely referenced in both academic and industry literature and provides best practices for routing, switching, and security. This project builds upon these established models by implementing them in a simulated environment.

Enterprise network design has been widely studied in both academic research and industry practice. One of the most commonly referenced approaches is the hierarchical network design model, which divides the network into three layers. This model improves scalability, performance, and fault isolation by clearly defining the route of each layer within the network infrastructure (Cisco Systems, 2023)

Oppenheimer (2011) emphasizes that top down network design begins with understanding business and technical requirements before selecting technologies. This approach ensures that network architecture aligns with organizational needs such as security and performance. Many modern enterprise networks adopt this methodology to reduce complexity and support future growth.

This builds upon these established design principles by implementing a hierarchical enterprise network within a simulated environment. Rather than proposing new architecture, this integrates proven design models and evaluates their effectiveness through simulation and testing.

2.2 VLAN Based Network Segmentation

VLANsegmentation plays a critical role in improving both network performance and security within enterprise environments. By dividing the network into separate logical groups, broadcast traffic is reduced and sensitive departments are isolated from unauthorized access. In this project, VLANs were implemented to separate departments such as HR, Sales, IT, and Management ensuring communication and enhanced security.

Virtual Local Area Networks (VLANs) are a fundamental technique used to logically segment networks without requiring separate physical infrastructure. VLANs reduce broadcast traffic and improve security by isolating groups of devices based on organizational roles (Kurose & Ross, 2021).

Previous research highlights VLAN segmentation as an effective method for protecting sensitive departments such as Human Resources and Finance from unauthorized access. Cisco documentation recommends VLANs as a best practice for enterprise environments due to their flexibility and scalability (Cisco Systems, 2022).

This project applies VLAN segmentation to separate departments within the enterprise network. By assigning each department to its own VLAN, the project demonstrates how logical segmentation can reduce security risks while maintaining efficient communication.

In addition to improving basic network organization, VLANs play a significant role in enforcing security policies within enterprise environments. By separating devices into logically defined groups, VLANs limit the ability of unauthorized users to access sensitive network resources. This isolation is especially important for departments that handle confidential information such as HR or management where unrestricted access could lead to data exposure or compliance violations (Stallings, 2020)

Research and industry case studies indicate that VLAN implementation can also improve overall network performance by reducing unauthorized broadcast traffic. When broadcast domains are confined to individual VLANs, network congestion is minimized, resulting in improved efficiency and responsiveness (Kurose & Ross 2021). This benefit becomes increasingly important as enterprise networks scale and support a growing number of devices and users.

From an administrative perspective, VLANs provide flexibility and ease of management. Network administrators can reorganize departments, add new users, or modify access policies without changing the physical network infrastructure. Cisco documentation highlights VLANs as a cost-effective solution for organizations that require scalable and adaptable network designs (Cisco Systems, 2022).

In this project, VLAN based segmentation is used not only as a structural component but also as a foundation for enforcing security controls. Each department within the simulated enterprise network is assigned to a dedicated VLAN, and inter-VLAN communication is carefully regulated using access control mechanisms. This approach demonstrates how VLANs can be combined with routing and security policies to create a secure and manageable network architecture.

2.3 Access Control and Network Security

Access Control Lists (ACLs) are a fundamental security mechanism used to control traffic flow within enterprise networks. Prior research highlights ACLs as a lightweight but effective method for enforcing security policies when properly configured.

This project builds on existing security practices by implementing ACLs to restrict inter department communication while allowing necessary access demonstrating how policy based security can be enforced at the network level.

Network security remains a critical concern for enterprise environments. Access Control Lists (ACLs) are widely used to filter network traffic based on predefined rules, allowing administrators to enforce security policies at the network layer (Stallings, 2020).

According to the National Institute of Standards and Technology (NIST), properly configured ACLs and firewall policies can significantly reduce the risk of unauthorized access and internal threats (Scarfone & Hoffman 2009). While ACLs are considered a basic security mechanism, they remain highly effective when combined with network segmentation techniques such as VLANs.

This project builds on existing security practices by implementing ACLs to control inter-VLAN communication. The use of ACLs ensures that departments can only access resources necessary for their functions, reinforcing the principle of least privilege.

Beyond basic traffic filtering, access control mechanisms play a crucial role in enforcing organizational security policies within enterprise networks. Access Control Lists (ACLs) allow network administrators to define explicit rules that permit or deny traffic based on factors such as IP address, protocol type, and port number. When implemented correctly, ACLs help prevent unauthorized access and reduce the attack surface of internal networks (Stallings 2020).

Industry standards emphasize the principle of least privilege, which states that users and systems should only be granted a minimum level of access required to perform their tasks. According to guidelines published by the National Institute of Standards and Technology (NIST), enforcing least privilege through network level controls significantly reduces the risk of both external and internal security threats (Scarfone & Hoffman 2009). ACLs are commonly used to enforce this principle by limiting communication between departments that do not require direct interaction.

Access control is particularly important in environments where sensitive data is distributed across multiple departments. Without proper restrictions, internal threats such as accidental data exposure or malicious insider activity can compromise organizational security. Research shows that combining logical network segmentation with access control mechanisms provides a layered defense strategy that is more effective than relying on a single security measure (Kurose & Ross 2021).

Access control policies are implemented to regulate inter-VLAN communication within the enterprise network. Specific ACL rules are configured to allow necessary services while blocking unauthorized access between departments. This approach demonstrates how network level security controls can be used to enforce organizational policies that improve security posture and maintain controlled communication across the enterprise network.

2.4 Network Simulation Tools

Network simulation tools such as Cisco Packet Tracer and GNS3are commonly used in academic settings to model real-world networking scenarios. Studies have shown that simulation based learning improves understanding of complex networking concepts without the risks associated with live network experimentation.

This leverages Cisco Packet Tracer to design, configure, and test an enterprise network in a controlled environment. This allows for safe and cost effective evaluation of network behavior. Network simulation tools are widely used in academic environments to model and test network designs without physical hardware. Cisco Packet Tracer is one of the most commonly used simulation tools due to its ease of use and support for Cisco Networking concepts (Cisco Networking Academy, 2024).

Studies have shown that simulation based learning improves students' understanding of complex networking concepts such as routing, switching, and security. Packet Tracer allows users to visualize packet flow, test configurations, and troubleshoot errors in a controlled environment (Tanenbaum & Wetherall 2021).

This project leverages Cisco Packet Tracer to design, configure, and test an enterprise network in a controlled environment, enabling systematic validation of connectivity, segmentation, and access-control policies without requiring physical hardware. Packet Tracer provides a simulation platform that allows network configuration to be implemented without the need for physical hardware. This enables safe testing of routing. VLAN segmentation, and security policies while minimizing cost and risk. In this project, the tool was used to validate connectivity to enforce access control rules and demonstrate network functionality.

Network simulation tools have become an essential component of both academic research and practical network design. These tools allow network architects and students to model complex enterprise networks, test configurations, and analyze performance without deploying physical hardware. Simulation environments reduce cost, minimize risk, and enable rapid experimentation, making them particularly valuable during the early stages of network planning (Tanenbaum & Wetherall 2021).

One of the primary advantages of network simulation is the ability to evaluate network behavior under different scenarios. Administrators can simulate traffic loads and security policies to observe how the network responds. This capability is especially important for enterprise environments, where downtime or misconfiguration can result in significant operational and financial consequences (Kurose & Ross 2021).

Cisco Packet Tracer is widely used in educational settings due to its support for core networking concepts such as routing, switching, VLAN configuration, and access control. While Packet Tracer does not replicate all features of physical hardware, studies have shown that it provides sufficient realism for teaching and validating fundamental enterprise networking designs (Cisco Networking Academy 2024). Its visual interface also enhances understanding by allowing users to observe packet flow and troubleshoot configuration errors.

Despite its advantages, network simulation has certain limitations. Simulated environments may not fully capture real-world hardware constraints and advanced protocol behavior. However, for the purpose of conceptual design, security policy testing, and educational analysis that shows how simulation tools remain an effective and widely accepted approach. This project acknowledges these limitations while leveraging simulation to demonstrate enterprise network design principles in a controlled environment.

2.5 Prior Work by the Author

The coursework and laboratory exercises in computer networking include basic routing and switching and being able to understand subnetting. This extends prior academic experience by integrating multiple networking concepts into a single comprehensive enterprise network design.

This shows how completed foundational coursework in computer networking, including routing, switching, subnetting, and basic network security. Laboratory exercises using Cisco Packet Tracer have provided hands-on experience with configuring network devices and troubleshooting connectivity issues.

This project extends prior academic work by integrating multiple networking concepts into a comprehensive enterprise network design. The project represents a progression from isolated laboratory exercises to a complete real-world network simulation.

2.6 Summary of Prior Related Work

The reviewed literature demonstrates a strong consensus regarding the importance of structured enterprise network design and logical segmentation. Industry best practices and academic research consistently emphasize hierarchical design models, VLAN based segmentation, and policy driven access control as foundational components of secure and scalable enterprise networks.

Previous studies and industry guidelines also support the use of network simulation tools as a valid method for designing and evaluating network architectures. Simulation enables cost-effective testing and provides valuable insights into network behavior and security before physical deployment.

Building upon this body of work integrates established enterprise networking techniques into a single simulated environment. By combining VLAN segmentation, access control policies, and network simulation will demonstrate a practical application of prior research while reinforcing best practices in enterprise network design.

3. Project Design and Analysis

3.1 System Architecture Overview

The proposed enterprise network was designed using a hierarchical structure consisting of an access layer and a routing layer. Two Layer 2 switches were deployed to connect end-user devices across multiple departments, while a central router was configured using a router-on-a-stick approach to enable inter-VLAN routing. An additional router was implemented to simulate an external Internet Service Provider (ISP) for testing external connectivity through Network Address Translation (NAT).

The network topology includes departmental VLAN segmentation for Human Resources, Sales, IT, Management, Guest Access, and internal servers. Trunk links were configured between switches and the router to allow multiple VLANs to traverse a single physical interface using IEEE 802.1Q encapsulation.

This architecture ensures scalability, logical segmentation, and centralized routing control while maintaining secure and controlled communication between departments.

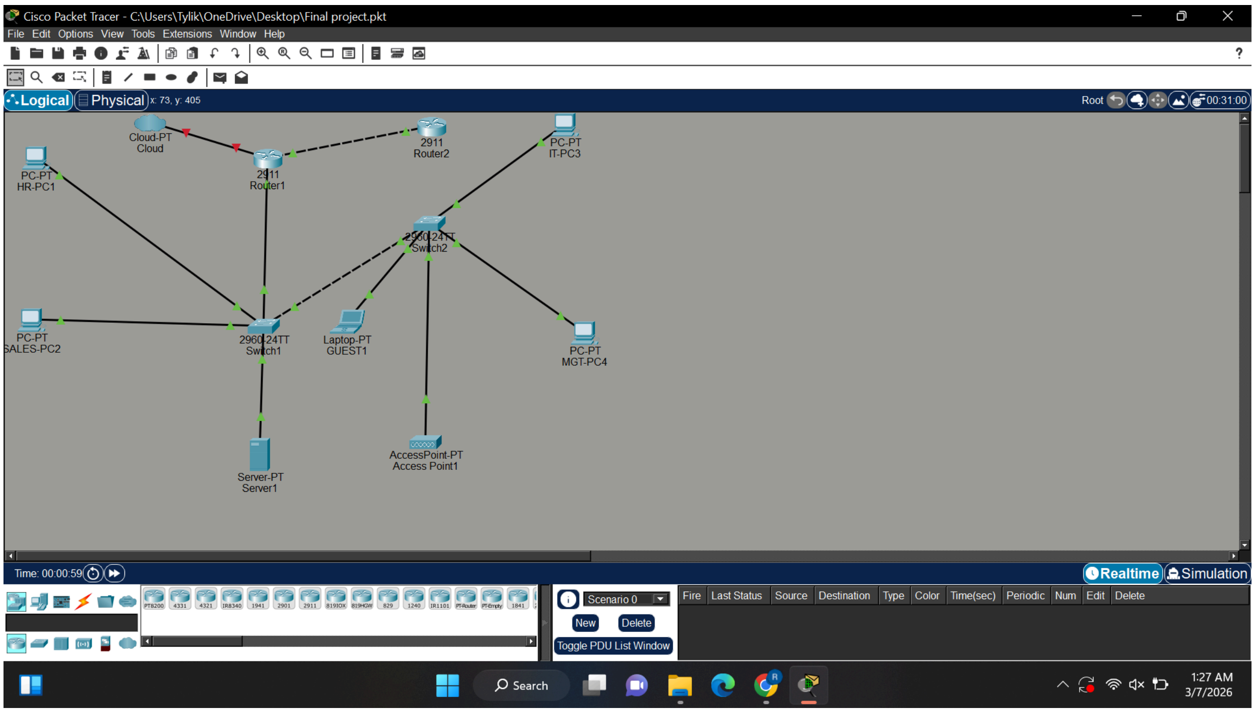

Figure 1. Final enterprise network topology with VLAN segmentation, router-on-a-stick configuration, and simulated ISP connection.

3.2 VLAN and IP Addressing Design

Each department was assigned a unique VLAN and subnet to logically separate broadcast domains and improve security. This segmentation reduces unnecessary traffic and ensures controlled communication between departments. The following VLAN structure was implemented:

VLAN	Department	Subnet
10	HR	192.168.10.0/24
20	Sales	192.168.20.0/24
30	IT	192.168.30.0/24
40	Management	192.168.40.0/24
50	Guest	192.168.50.0/24
60	Servers	192.168.60.0/24

Each VLAN uses a /24 subnet for simplicity and ease of management. The router subinterfaces were configured as the default gateway for each VLAN, enabling inter VLAN communication.

Switch trunking was configured to allow VLAN traffic between switches and the router using IEEE 802.1Q encapsulation.

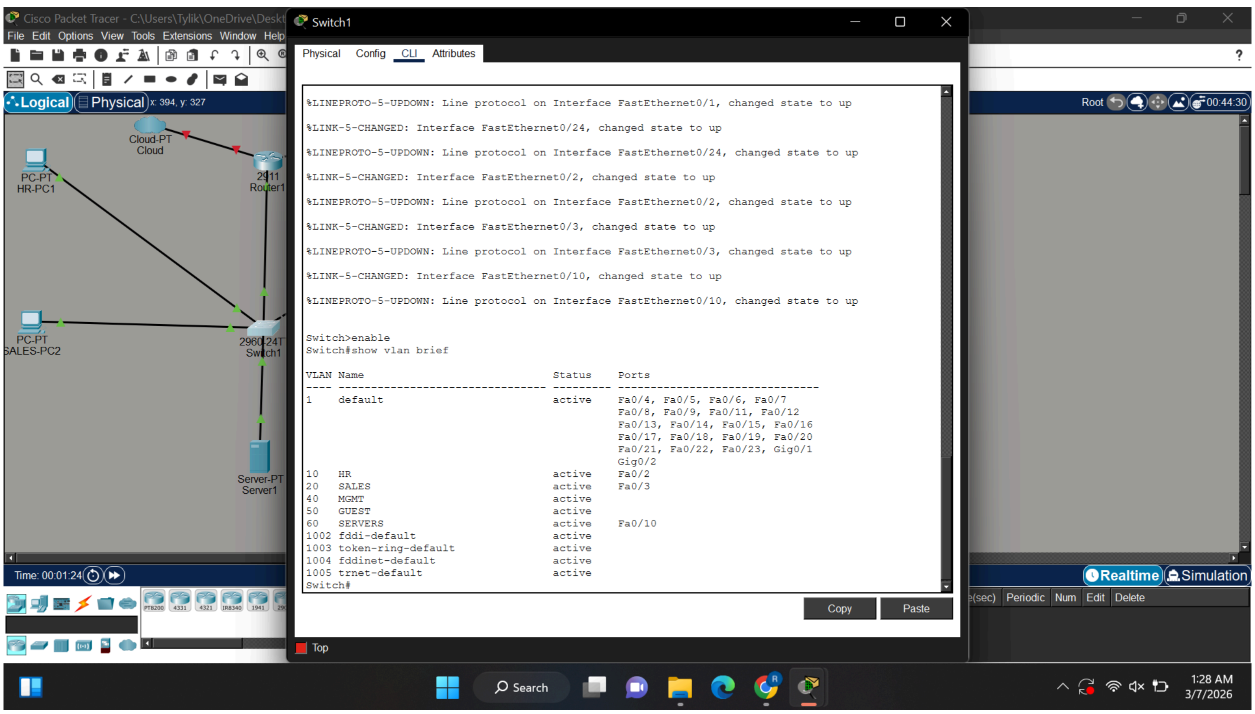

Figure 2. VLAN configuration and port assignments on Switch 1.

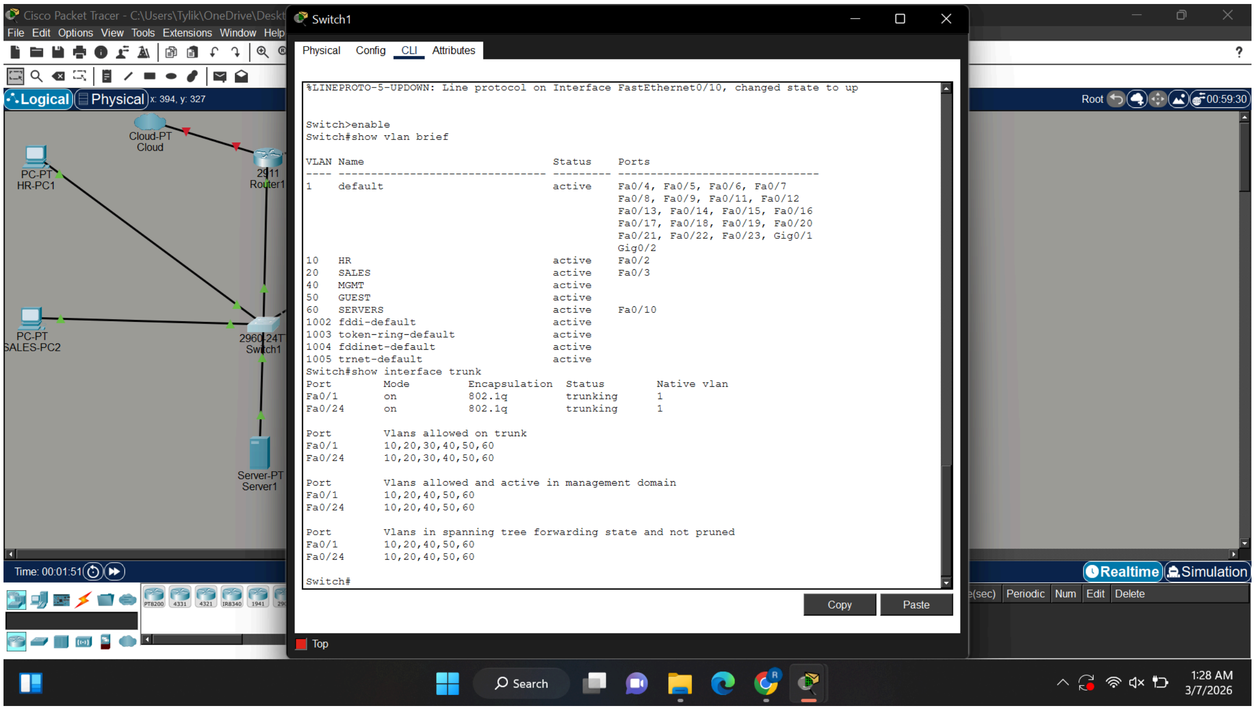

Figure 3.Trunk configuration allowing VLAN traffic between switches and router.

In designing the VLAN structure, logical separation was prioritized to reduce broadcast traffic and improve security boundaries between departments. A /24 subnet was selected for each VLAN to simplify addressing, ensure scalability for future device growth, and maintain consistency across departmental segments. Assigning a dedicated gateway address within each subnet allows centralized routing control while preserving isolation at the Layer 2 level. This approach reflects industry best practices for enterprise segmentation and enhances both performance and manageability.

3.3 Routing Design (Router-on-a-Stick)

Inter-VLAN routing was implemented using a router-on-a-stick configuration. A single physical interface on the router was divided into multiple logical subinterfaces using 802.1Q encapsulation. Each subinterface corresponds to a specific VLAN and was assigned a gateway IP address for that subnet. This approach allows efficient inter-VLAN communication without requiring a Layer 3 switch. The router performs packet forwarding between VLANs while maintaining logical segmentation.

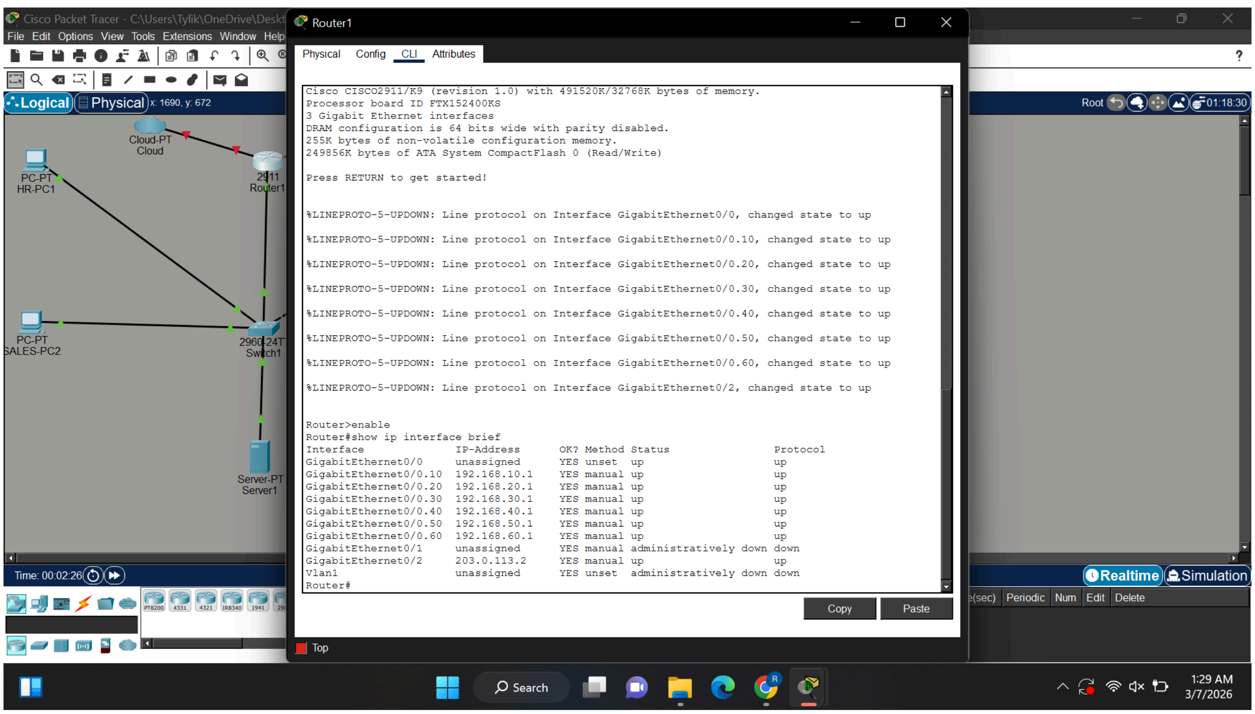

Figure 4. Router subinterfaces configured for inter-VLAN routing.

The router-on-a-stick approach was selected due to its efficiency and suitability for small-to-medium enterprise environments. By using IEEE 802.1Q encapsulation, a single physical interface on the router can service multiple VLANs through logical subinterfaces. This design reduces hardware requirements while maintaining full inter-VLAN routing capabilities. Although Layer 3 switches could also perform this function, the router-based solution was appropriate for this simulated enterprise environment and allowed clear demonstration of VLAN tagging and routing processes.

3.4 Security Policy Implementation

1. HR Protection Policy:
    a. Only Management (VLAN 40) is allowed to access HR resources (VLAN 10). Sales and other departments are denied access.
2. Guest Isolation Policy
    a. Guest VLAN users are restricted from accessing any internal VLANs, including HR, Sales, IT, Management, and Servers.

Two primary security policies were implemented. First, the HR Protection Policy ensures that only Management (VLAN 40) can access HR resources (VLAN 10), while Sales and other departments are denied access. Second, the Guest Isolation Policy restricts guest VLAN users from accessing any internal VLANs, including HR, Sales, IT, Management, and Servers.

Extended ACLs were applied as close to the source as possible to follow Cisco best practices.

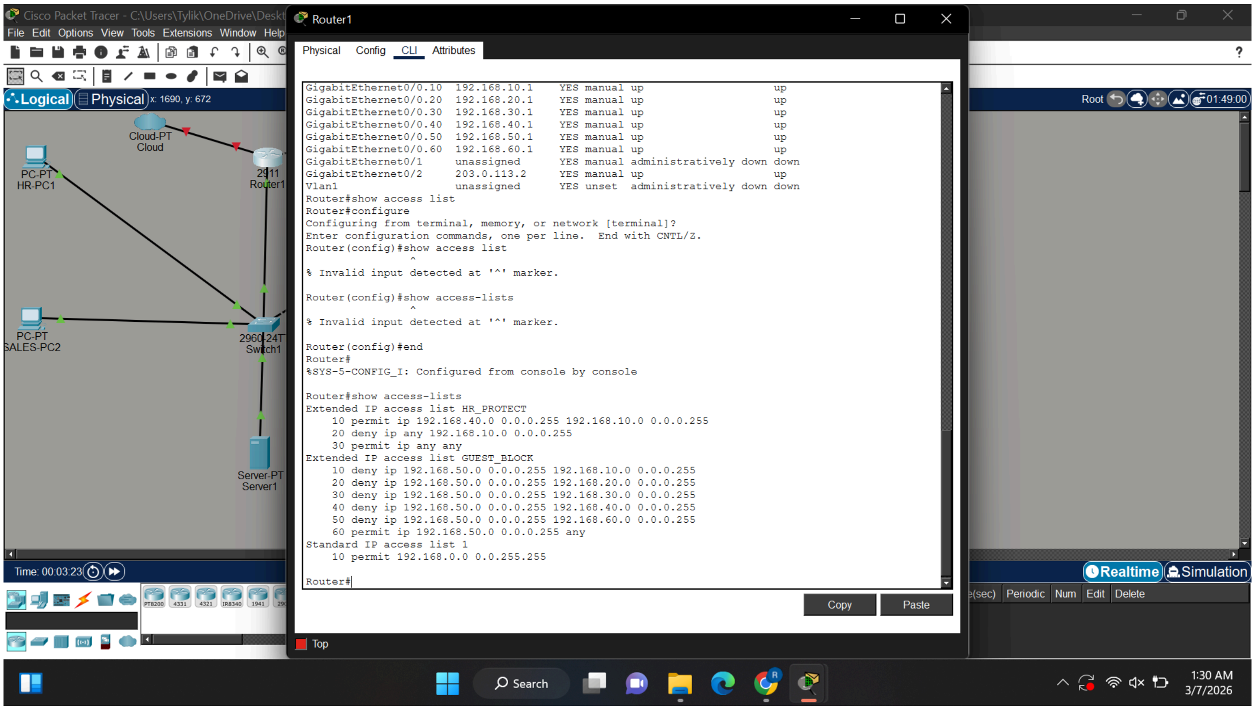

Figure 5. Extended ACLs enforcing HR protection and guest isolation.

Access control policies were implemented based on the principle of least privilege, ensuring that departments only access resources necessary for their functions. Extended ACLs were placed as close to the source of traffic as possible, following Cisco best practices for traffic filtering. By strategically applying ACLs on specific VLAN subinterfaces, unauthorized communication was successfully blocked while legitimate traffic remained unaffected. This layered security approach enhances internal threat mitigation and demonstrates practical enterprise policy enforcement.

3.5 Network Address Translation (NAT)

To enable internal devices using private IP addresses to access external networks, NAT overload (PAT) was implemented on the edge router. The router translates multiple internal private addresses into a single public IP address assigned to the WAN interface. The configuration designates VLAN subinterfaces as inside interfaces and the WAN interface as outside. When internal hosts communicate with the ISP router, their private addresses are translated to the router's public IP address.

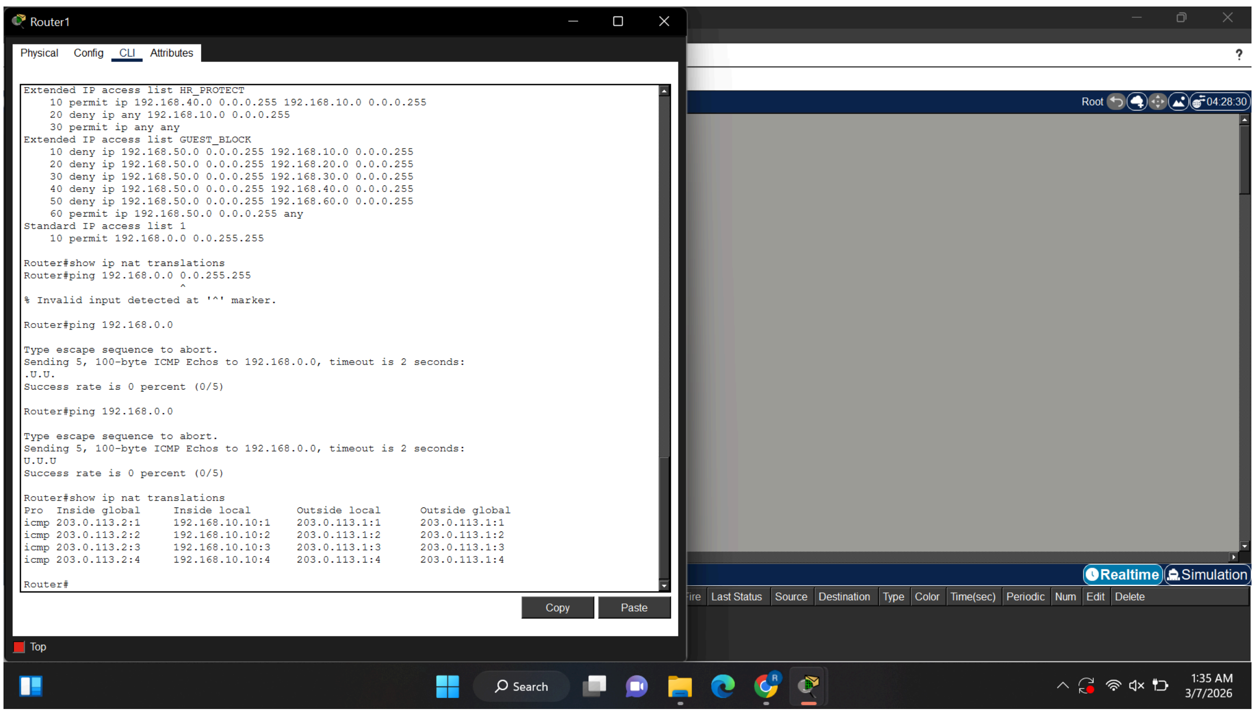

Figure 6. NAT translation table showing internal private addresses mapped to the public WAN interface.

Network Address Translation was implemented using Port Address Translation (PAT), allowing multiple internal private IP addresses to share a single public WAN address. This configuration reflects real-world enterprise edge router deployments, where internal addressing schemes must be translated before communicating with external networks. The successful NAT translation entries confirm that internal VLANs are able to access external resources while preserving private IP addressing internally.

3.6 Testing and Validation

A series of tests were conducted to validate routing, security policies, and external connectivity within the enterprise network. These tests were designed to ensure that each configured component behaved according to the intended design. Both successful and restricted communication scenarios were evaluated to verify that routing functionality, access control rules, and NAT configuration were working correctly.

Test Case	Expected Result	Actual Result
HR → Server	Allowed	Successful
Sales → HR	Blocked	Blocked
Management → HR	Allowed	Successful
Guest → Internal VLANs	Blocked	Blocked
HR → ISP	Allowed	Successful
Guest → ISP	Allowed	Successful

The results confirm that VLAN segmentation, routing, ACL enforcement, and NAT functionality operate as intended.

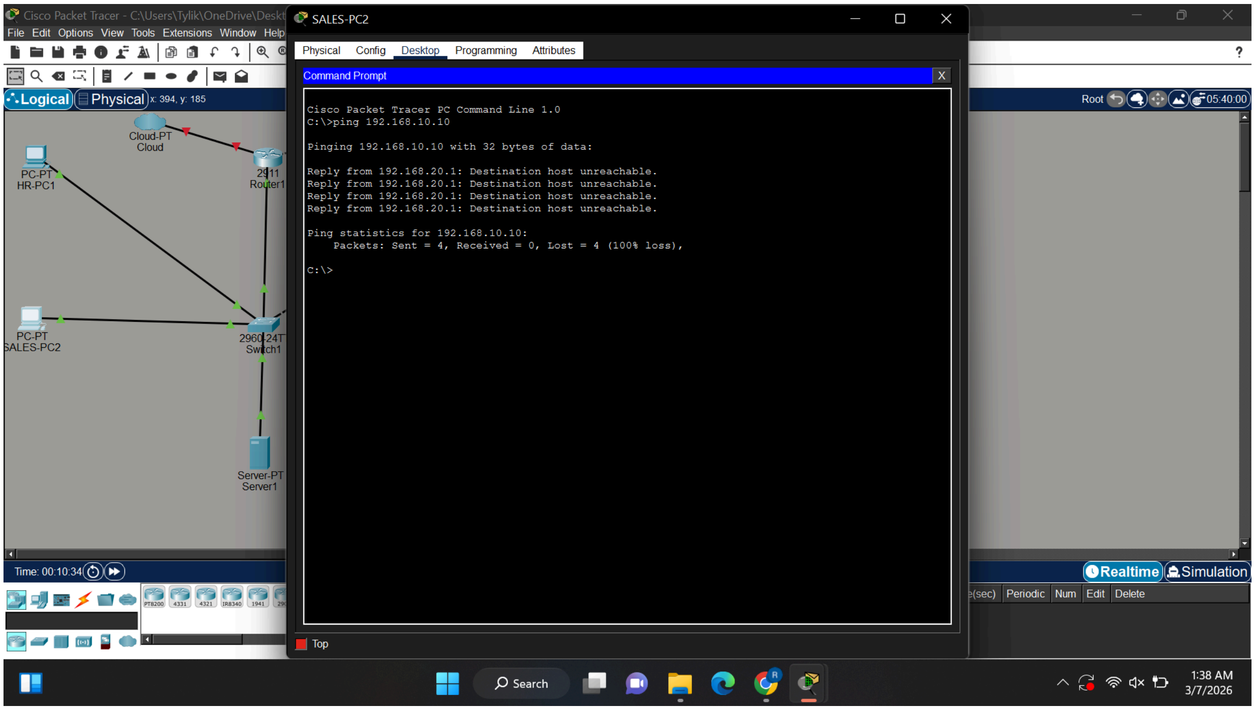

Figure 7. Blocked access from Sales VLAN to HR VLAN demonstrating ACL enforcement.

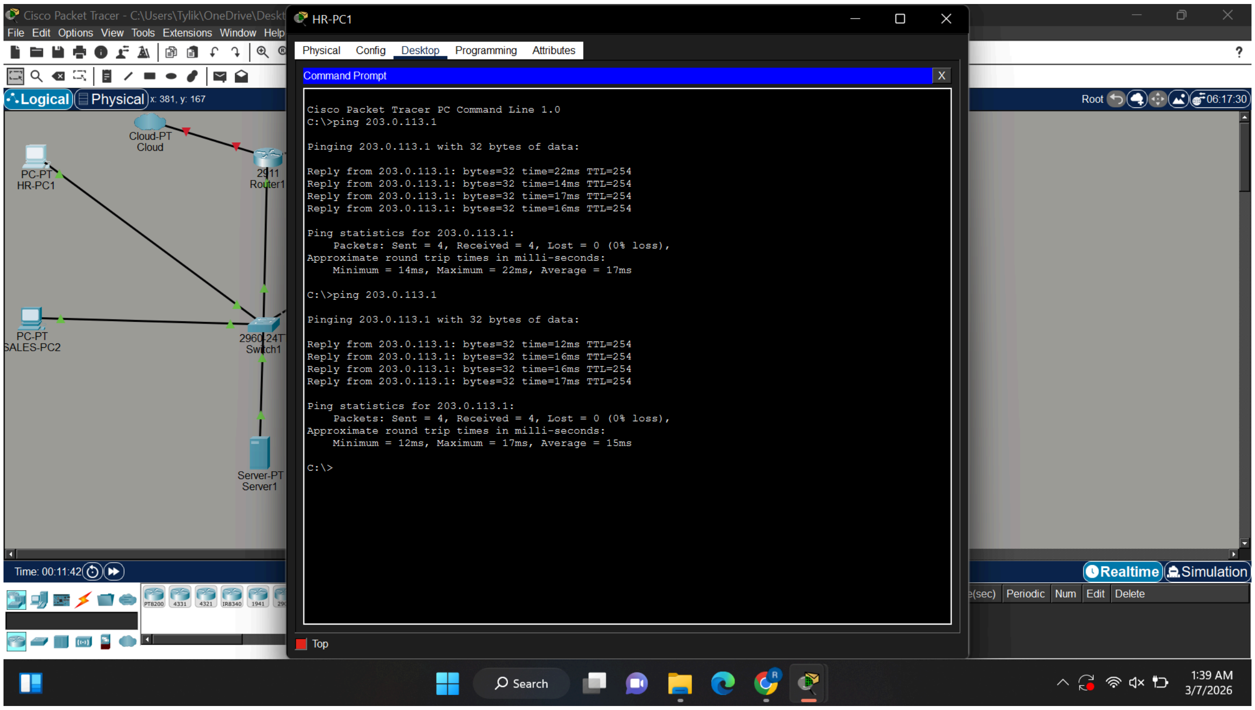

Figure 8. Successful external connectivity through NAT configuration.

3.7 Analysis and Discussion

The implementation of the proposed enterprise network successfully demonstrated the effectiveness of VLAN segmentation, router-on-a-stick inter-VLAN routing, access control enforcement, and NAT configuration. All functional requirements were validated through structured testing procedures. The results confirmed that unauthorized inter-department communication was blocked while authorized traffic was permitted, aligning with the intended security policies.

One of the key strengths of this design is its scalability and modular structure. Additional VLANs and departments can be incorporated without significant architectural changes. Furthermore, the use of simulation tools allowed comprehensive testing without the cost and complexity of physical deployment, making the solution practical for educational and small enterprise environments.

However, certain limitations exist within the simulation environment. Packet Tracer does not fully emulate advanced hardware performance characteristics, latency, or complex routing protocol behaviors. Future improvements could include implementing dynamic routing protocols such as OSPF, adding firewall devices for deeper inspection, or migrating the design to a virtual lab environment such as GNS3 for enhanced realism. Overall, the project demonstrates a secure, scalable, and well-structured enterprise network design that aligns with industry best practices.

3.8 Design Decisions and Justification

The design of the enterprise network was guided by principles of scalability, security, and simplicity. VLAN segmentation was selected to logically separate departments without requiring additional physical hardware. This approach allows flexible network management and improves security by isolating sensitive areas such as the HR and Management VLANs. Router-on-a-stick was chosen as the inter-VLAN routing method due to its efficiency and suitability for simulated environments. While Layer 3 switches could provide higher performance, the router-based solution offers a clear demonstration of VLAN tagging and routing concepts.

Extended ACLs were implemented to enforce role-based access control policies. This approach ensures that only authorized users can access sensitive resources, aligning with industry best practices such as the principle of least privilege. NAT was included to simulate real world internet connectivity, allowing internal devices to communicate with external networks. This demonstrates how enterprise networks handle private to public address translation in practical scenarios.

4. Implementation and Evaluation

4.1 System Implementation Overview

This project was implemented using Cisco Packet Tracer to simulate an enterprise network environment. The implementation included configuring VLAN segmentation, inter VLAN routing using router-on-a-stick, access control policies using extended ACLs, and Network Address Translation (NAT) for external connectivity. Each component was configured step by step and tested to ensure proper functionality and alignment with network design principles.

4.2 Functional Components

The following key functional components of the network were successfully implemented and validated through testing. These components demonstrate the core capabilities of the enterprise network, including segmentation, routing, security, and external connectivity.

- VLAN segmentation across multiple departments
- Inter-VLAN routing using router-on-a-stick
- Role-based access control using extended ACLs
- Guest network isolation from internal VLANs
- NAT configuration for internet connectivity
- Successful communication between authorized network segments

Each functional component was validated through structured testing scenarios to confirm that the network behaved as expected.These tests ensured that authorized communication was allowed while unauthorized access was properly restricted. The results of these tests demonstrate that the implemented design meets the intended security and performance requirements.

4.3 Test Cases and Results

The following test cases were developed to evaluate network functionality and verify that the system behaves to the defined use cases.

Test Case	Description	Expected Result	Actual Result
TC1	HR accesses server	Allowed	Successful
TC2	Sales → HR	Blocked	Blocked
TC3	Mgmt → HR	Allowed	Successful
TC4	Guest → Internal	Blocked	Blocked
TC5	HR → ISP	Allowed	Successful

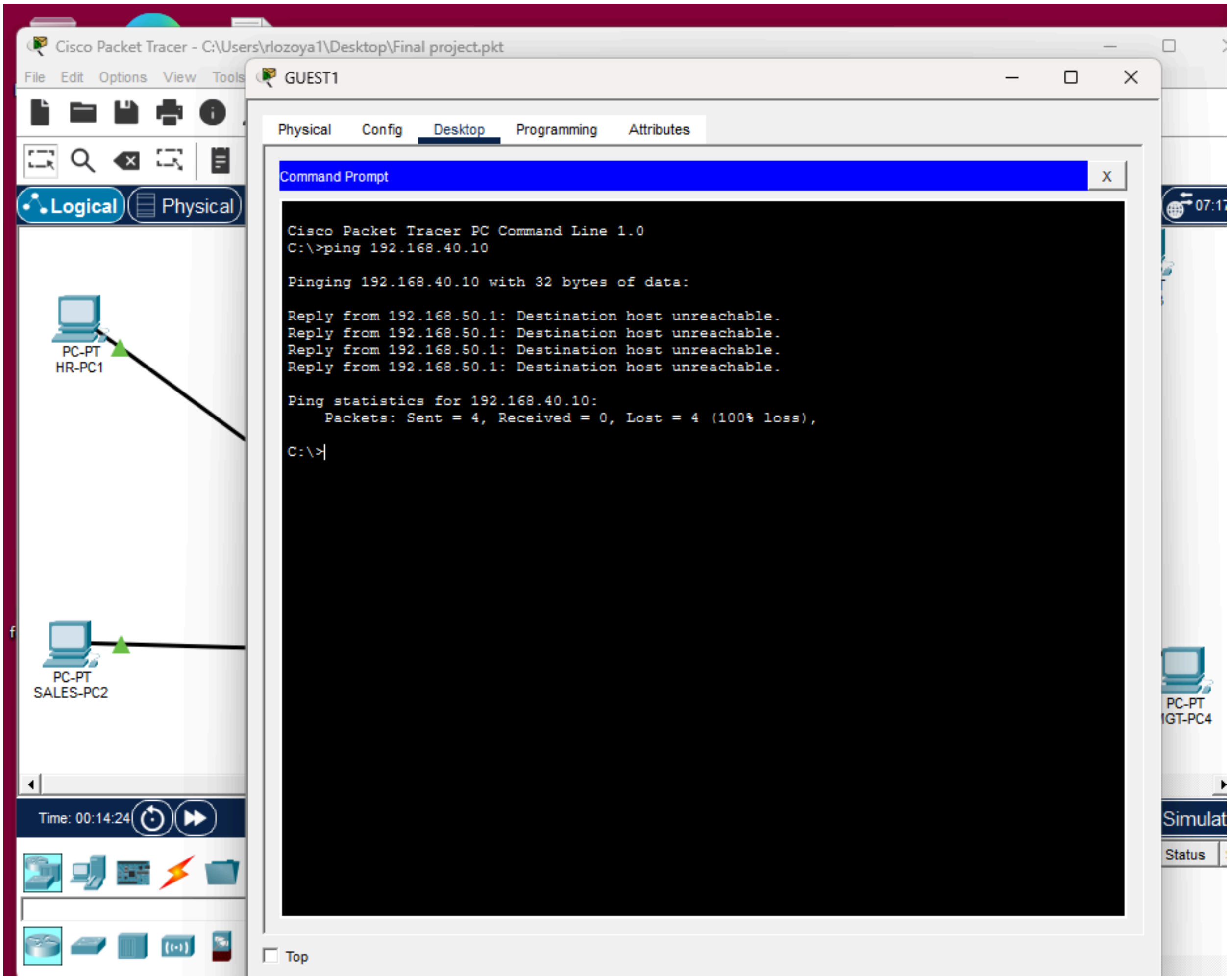

Figure 9. An example above shows guests not being able to ping management.

4.4 Challenges Encountered

Several challenges were encountered during the implementation of the network. One of the main challenges was correctly configuring trunk links and ensuring VLAN consistency across switches. Another issue involved properly applying ACLs on the correct interfaces, as incorrect placement initially allowed unintended traffic. Additionally, configuring NAT required careful setup of inside and outside interfaces to ensure proper internet connectivity. These challenges required troubleshooting using CLI commands and iterative testing.

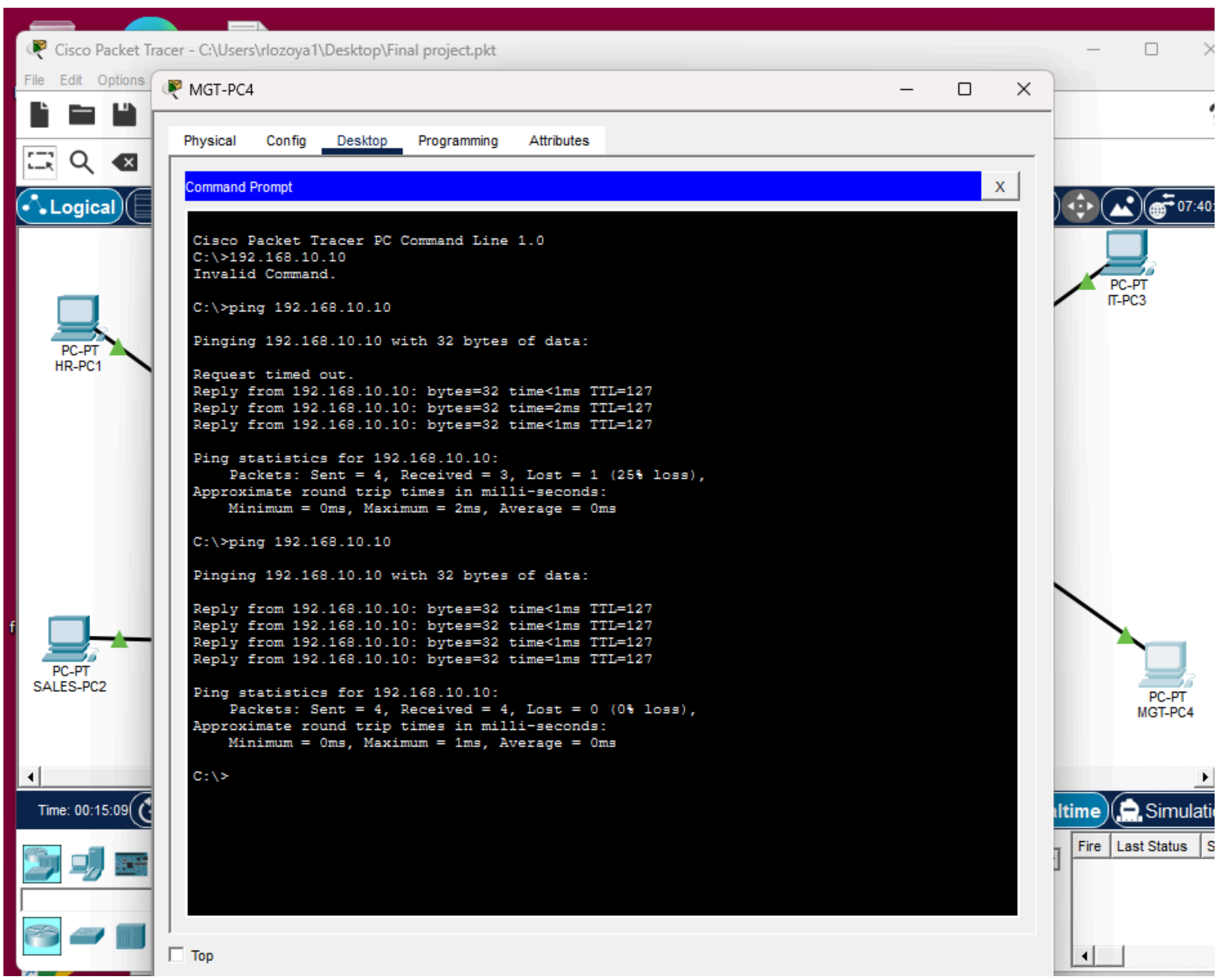

Figure 10. The image above shows ping not being able to go through at the first attempt.

4.5 Solutions and Improvements

To address these challenges, configurations were reviewed and corrected by verifying VLAN assignments, trunk links, and routing interfaces. ACL placement was adjusted to follow best practices by applying them close to the source of traffic. NAT configuration issues were resolved by correctly identifying inside and outside interfaces and verifying tables. These improvements ensured that the network behaved as expected and met all design requirements.

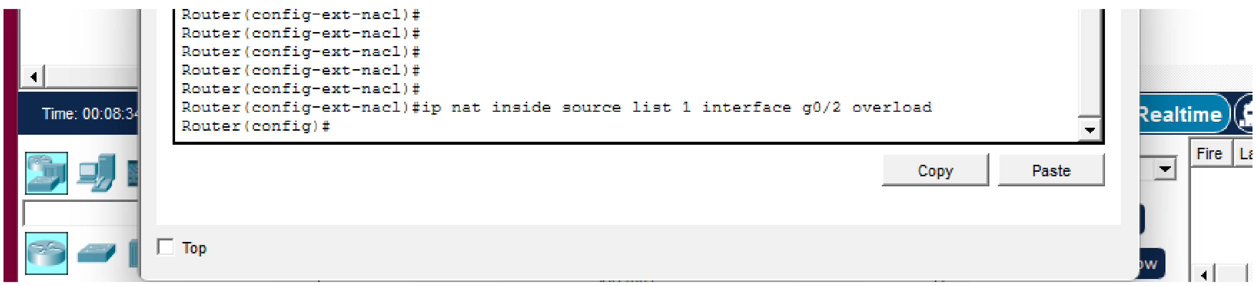

Figure 11. NAT configuration issues were resolved by correctly identifying inside and outside interfaces and verifying translation tables. NAT overload configuration allowing internal devices to access external networks.

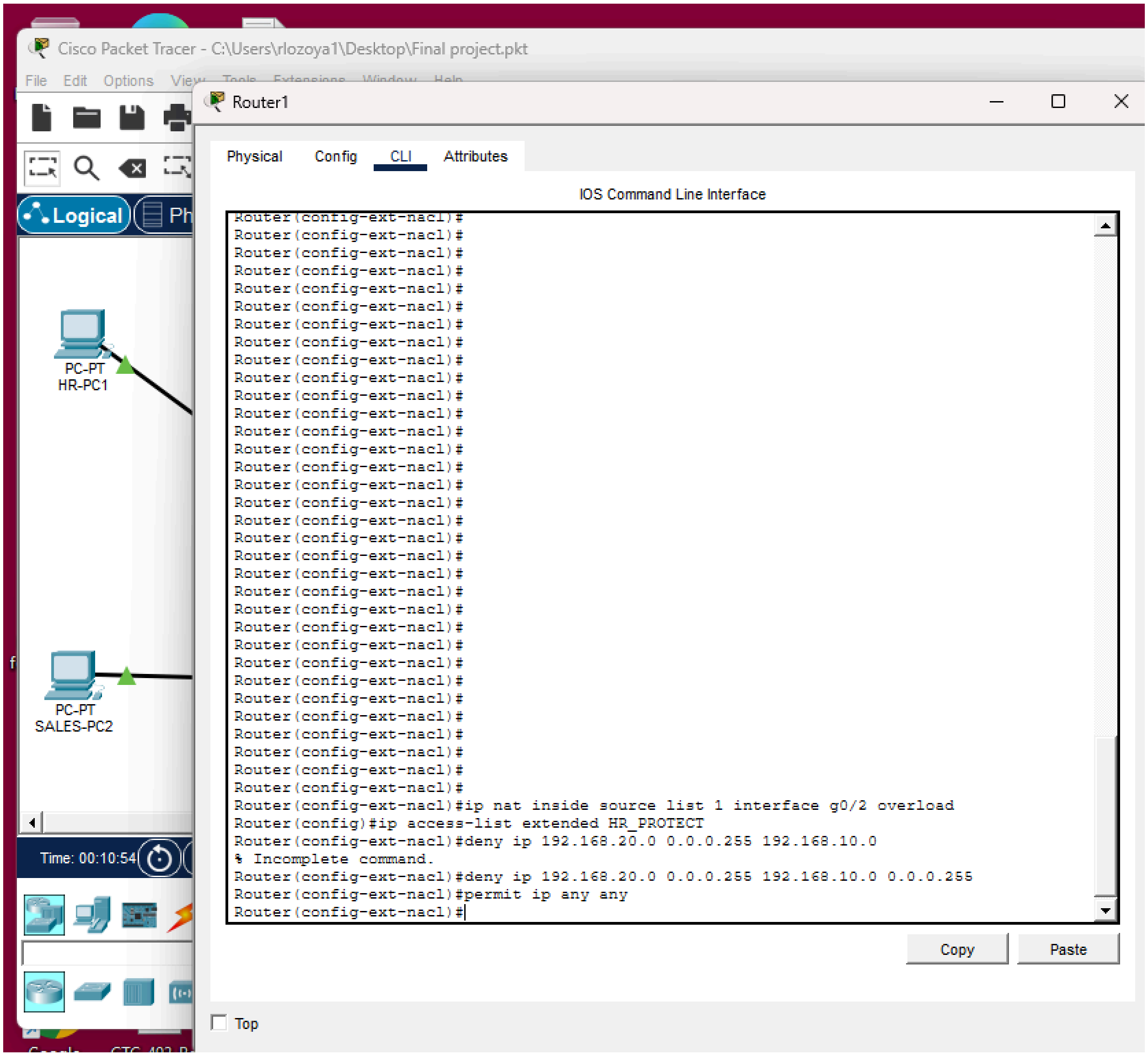

Figure 12. Extended ACL configuration used to restrict unauthorized access between VLANs.

4.6 Limitations and Future Work

Although the network was successfully implemented, there are limitations within the simulation environment. The simulation environment in Cisco Packet Tracer does not fully replicate the real world hardware behavior or advanced routing protocols. Future improvements for this project include implementing dynamic routing protocols such as OSPF to enhance scalability, as well as adding a firewall to provide more advanced security beyond basic ACLs. Additional testing and refinement of network configurations will also be performed to further validate performance and ensure the design aligns with real world enterprise networking standards.

4.7 Network Performance and Design Evaluation

The performance and behavior of the network were evaluated based on connectivity, security enforcement, and overall design efficiency. The implementation of VLAN segmentation significantly reduced unnecessary broadcast traffic and improved network organization by isolating departments into separate logical groups. This segmentation enhances both performances and security by limiting the scope of network communication.

Inter-VLAN routing using router-on-a-stick was effective in enabling communication between authorized VLANs while maintaining logical separation. Although this approach is suitable for small-to-medium networks, it may introduce limitations in scalability compared to Layer 3 switching in larger enterprise environments.

The implementation of access control lists (ACLs) demonstrated strong security enforcement by restricting unauthorized communication while allowing necessary access. This validates the effectiveness of role-based access control within the network. Additionally, NAT configuration allowed internal devices to communicate with external networks using a single public IP address, demonstrating real-world enterprise network functionality. Overall, the network design successfully meets the intended goals of security, scalability, and controlled communication.

Appendix A: GitHub Repository

The complete project files, including the Packet Tracer simulation and supporting documentation are available at the following GitHub repository:
https://github.com/rlozoya1/secure-enterprise-network

Resources and References

Resource	Purpose	Estimated Cost
Cisco Packet Tracer	Network simulation	Free
Personal computer	Development & testing	$0
Textbooks	Reference material	$0-$100
Cisco IOS documentation	Configuration reference	Free

Cisco Systems. (2023). *Enterprise network design guide*. Cisco Press.

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